Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel Soares & Sons Dairy

Physical address of dairy:

7018 Vivian RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X002-X055-XXXX

B. OPERATORS

Soares, Manuel

Operator name: Soares, Manuel

Telephone no.: (209) 537-5692 (Landline) (Cellular)

7018 Vivian RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Soares, Manuel

Legal owner name: Soares, Manuel

Telephone no.: (209) 537-5692 (Landline) (Cellular)

7018 Vivian RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1	230	360	245	140	0
Maximum number	1	230	360	245	140	0
Average number	1	230	360	245	140	0
Avg live weight (lbs)	1	1,300	900	600		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,609.07 *tons per reporting period*

Total nitrogen from manure: 106,773.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 74,741.10 *lbs per reporting period*

Total phosphorus from manure: 15,800.20 *lbs per reporting period*

Total potassium from manure: 146.18 *lbs per reporting period*

Total salt from manure: 53,359.35 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,997,000 *gallons*

Total nitrogen generated: 23,115.78 *lbs*

Total phosphorus generated: 5,173.61 *lbs*

Total potassium generated: 68,773.40 *lbs*

Total salt generated: 242,136.02 *lbs*

	4,752,000	*gallons applied*
+	1,245,000	*gallons exported*
-	0	*gallons imported*
=	5,997,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well #1	Ground water
Domestic Well - Well #2	Ground water
Domestic Well - Well #3	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
08/13/2023	Process wastewater	1,245,000.00 *gal*	647.00	198.00	0.00	1.50	105.00	1,310.00		2,900

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	6,737.61	1,090.90	13,610.28	30,129.62
Total exports for all materials	6,737.61	1,090.90	13,610.28	30,129.62

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy	76	76	2	process wastewater	X022-X002-X005-XXXX
W-N-S-C	248	248	1	manure	X017-X062-X031-XXXX
Totals for areas that were used for application	324	324	3		
Totals for areas that were not used for application					
Land application area totals	324	324	3		

B. CROPS AND HARVESTS

Dairy

Field name: Dairy

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	567.89 *ton*	Dry-weight		65.7	11,000.00	1,200.00	13,000.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.70	126.04	13.75	148.95	905.18

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/04/2023	889.88 *ton*	Dry-weight		70.2	16,500.00	3,900.00	16,100.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.17	257.38	60.84	251.15	1,123.13

W-N-S-C

W-N-S-C

Field name: W-N-S-C

01/15/2008: Almond, in shell

Crop: Almond, in shell Acres planted: 248 Plant date: 01/15/2008

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	297.62 *ton*	Dry-weight		7.3	18,400.00	1,700.00	19,500.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	130.00	20.00	100.00	0.00
Total actual harvest content	1.20	40.94	3.78	43.39	126.82

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy - 11/17/2022: Oats, silage-soft dough

Field name: Dairy

Crop: Oats, silage-soft dough — Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.69	18.56	302.45	1,363.82	756,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.28	4,445,100.00 *gal*
Application event totals		74.69	18.56	302.45	1,391.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.64	16.79	273.65	1,233.93	684,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.80	4,041,000.00 *gal*
Application event totals		67.64	16.79	273.65	1,258.73	

Dairy - 05/15/2023: Corn, silage

Field name: Dairy

Crop: Corn, silage — Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.58	4,983,900.00 *gal*
Application event totals		0.00	0.00	0.00	30.58	

Dairy - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	126.06	20.41	254.65	563.73	792,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.75	4,849,200.00 *gal*
Application event totals		126.06	20.41	254.65	593.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.58	4,983,900.00 *gal*
Application event totals		0.00	0.00	0.00	30.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.09	21.14	260.13	1,010.03	828,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.37	4,950,225.00 *gal*
Application event totals		70.09	21.14	260.13	1,040.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.09	21.14	260.13	1,010.03	828,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.75	4,849,200.00 *gal*
Application event totals		70.09	21.14	260.13	1,039.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.14	22.05	271.44	1,053.94	864,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.99	5,051,250.00 *gal*
Application event totals		73.14	22.05	271.44	1,084.94	

Dairy - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.37	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	30.37	

W-N-S-C - 01/15/2008: Almond, in shell

Field name: W-N-S-C

Crop: Almond, in shell

Plant date: 01/15/2008

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.82	29,499,300.00 *gal*
Application event totals		0.00	0.00	0.00	24.82	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.65	29,297,250.00 *gal*
Application event totals		0.00	0.00	0.00	24.65	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.65	29,297,250.00 *gal*
Application event totals		0.00	0.00	0.00	24.65	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.73	29,398,275.00 *gal*
Application event totals		0.00	0.00	0.00	24.73	

W-N-S-C - 01/15/2008: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.22	28,792,125.00 *gal*
Application event totals		0.00	0.00	0.00	24.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.31	28,893,150.00 *gal*
Application event totals		0.00	0.00	0.00	24.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.65	29,297,250.00 *gal*
Application event totals		0.00	0.00	0.00	24.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.48	29,095,200.00 *gal*
Application event totals		0.00	0.00	0.00	24.48	

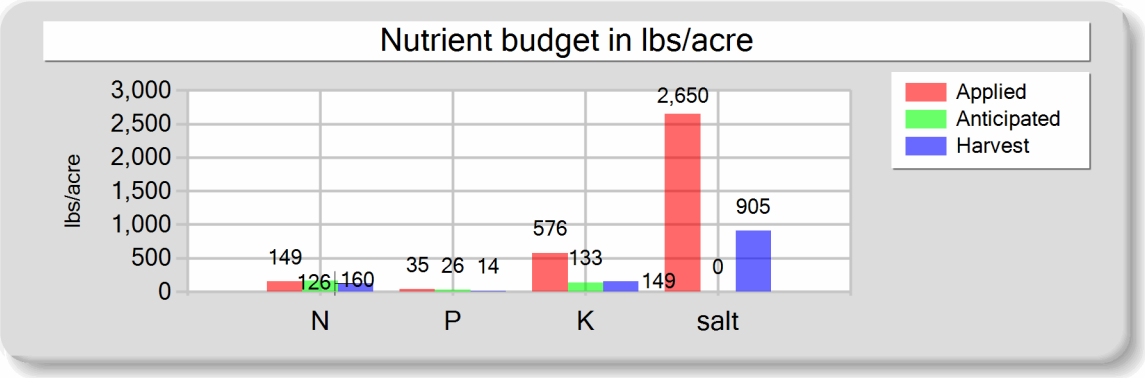
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	30.20	14.11	68.72	0.00	250.00 *ton*
Application event totals		30.20	14.11	68.72	0.00	

B. NUTRIENT BUDGET

Dairy - 11/17/2022: Oats, silage-soft dough

Field name: Dairy | Crop: Oats, silage-soft dough | Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	142.32	35.34	576.10	2,597.75
Fresh water	0.00	0.00	0.00	52.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.32	35.34	576.10	2,649.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.04	13.75	148.95	905.18
Nutrient balance	23.29	21.59	427.15	1,744.64
Applied to removed ratio	1.18	2.57	3.87	2.93

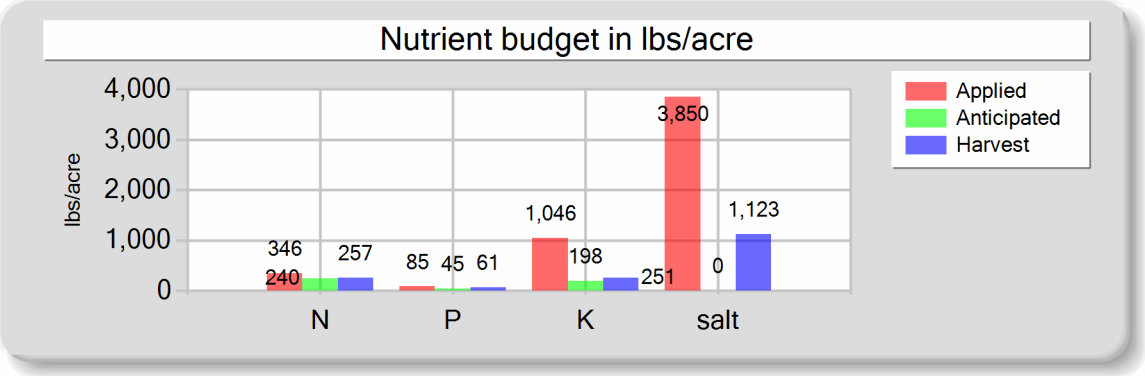
Fresh water applied	
8,486,100.00	*gallons*
312.51	*acre-inches*
9.19	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy - 05/15/2023: Corn, silage

Field name: Dairy | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	339.39	84.74	1,046.35	3,637.73
Fresh water	0.00	0.00	0.00	212.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	346.39	84.74	1,046.35	3,850.15
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	257.38	60.84	251.15	1,123.13
Nutrient balance	89.00	23.90	795.20	2,727.02
Applied to removed ratio	1.35	1.39	4.17	3.43

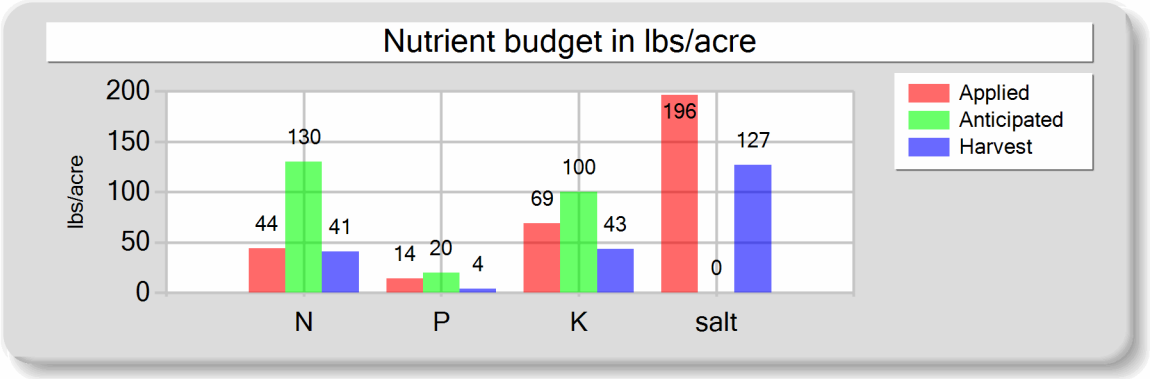
Fresh water applied	
34,617,900.00	*gallons*
1,274.86	*acre-inches*
37.50	*inches/acre*

Process wastewater applied	
3,312,000.00	*gallons*
121.97	*acre-inches*
3.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

W-N-S-C - 01/15/2008: Almond, in shell

Field name: W-N-S-C Crop: Almond, in shell Plant date: 01/15/2008



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	30.20	14.11	68.72	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	196.49
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	44.20	14.11	68.72	196.49
Anticipated crop nutrient removal	130.00	20.00	100.00	0.00
Actual crop nutrient removal	40.94	3.78	43.39	126.82
Nutrient balance	3.26	10.32	25.34	69.66
Applied to removed ratio	1.08	3.73	1.58	1.55

Fresh water applied	
233,569,800.00	*gallons*
8,601.58	*acre-inches*
34.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	7,800.00	38,000.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure
Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 3.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	4,900.00	27,200.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	647.00	198.00	0.00	1.50	105.00	1,310.00								9,010.00	2,900
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	221.00	0.00	0.90	104.00	1,280.00								8,570.00	4,970
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	301.00	177.00	0.00	0.50	96.10	1,130.00								7,400.00	6,050
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	402.00	229.00	0.00	1.50	100.00	1,630.00								9,100.00	7,350
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well #1

Domestic Well - Well #1

Well #1

Sample description: Well #1

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			24.30								1,060.00	
DL			0.10								10.00	

Domestic Well - Well #2

Well #2

Sample description: Well #2

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			26.10								1,140.00	
DL			0.10								10.00	

Domestic Well - Well #3

Well #3

Sample description: Well #3

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			27.50								1,480.00	
DL			0.10								10.00	

Irrigation Water - Canal

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								29.50	25
DL	0.50	0.50	0.50								10.00	10

D. SOIL ANALYSES

W-N-S-C

WNSC 1

Sample and source description: WNSC 1

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			238.00				
DL			1.00				

WNSC 2

Sample and source description: WNSC 2

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			237.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

Dairy - 11/17/2022: Oats, silage-soft dough

Dairy - 11/17/2022: Oats, silage-soft dough

Dairy

Sample and source description: Dairy

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,200.00	13,000.00		7.90
DL	100.00	100.00	100.00		0.01

Dairy - 05/15/2023: Corn, silage

Dairy

Sample and source description: Dairy

Sample date: 09/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,900.00	16,100.00		7.20
DL	100.00	100.00	100.00		0.01

W-N-S-C - 01/15/2008: Almond, in shell

W-N-S-C #1

Sample and source description: W-N-S-C #1

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	2,300.00	17,300.00		4.90
DL	100.00	100.00	100.00		0.01

W-N-S-C - 01/15/2008: Almond, in shell

W-N-S-C #2

Sample and source description: W-N-S-C #2

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,000.00	21,000.00		5.80
DL	100.00	100.00	100.00		0.01

W-N-S-C #3

Sample and source description: W-N-S-C #3

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	1,700.00	19,500.00		5.70
DL	100.00	100.00	100.00		0.01

W-N-S-C #4

Sample and source description: W-N-S-C #4

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,000.00	2,400.00	18,200.00		5.60
DL	100.00	100.00	100.00		0.01

W-N-S-C - 01/15/2008: Almond, in shell

W-N-S-C #5

Sample and source description: W-N-S-C #5

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	2,400.00	15,900.00		46.00
DL	100.00	100.00	100.00		0.01

W-N-S-C #6

Sample and source description: W-N-S-C #6

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,700.00	2,600.00	15,800.00		4.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

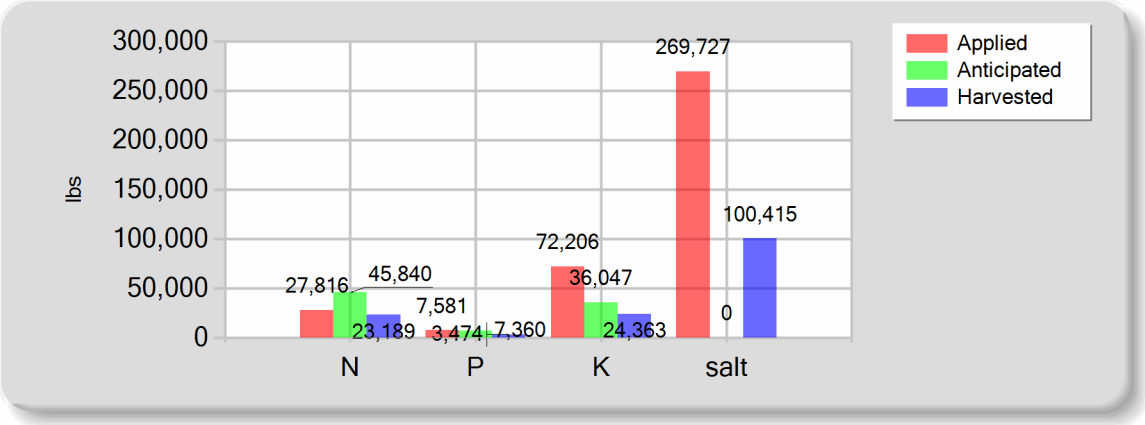
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

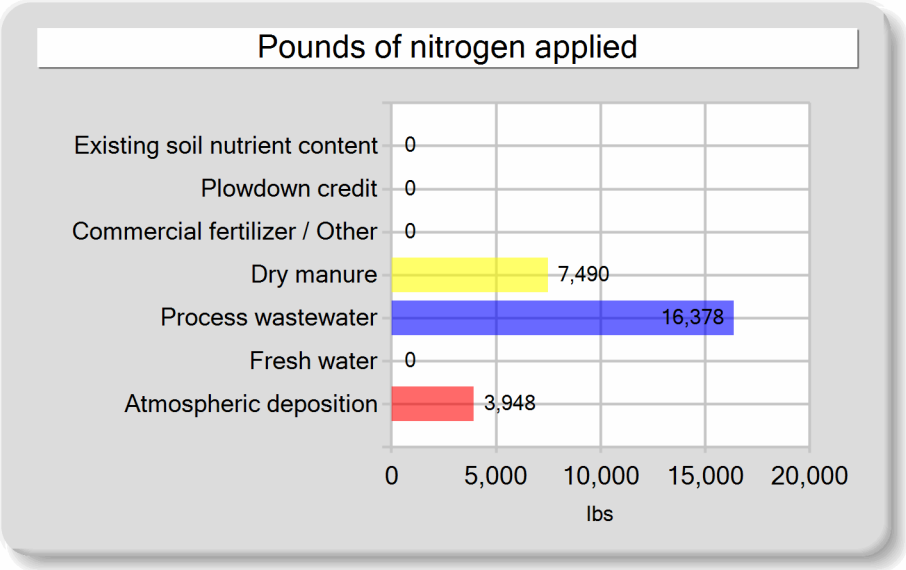
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

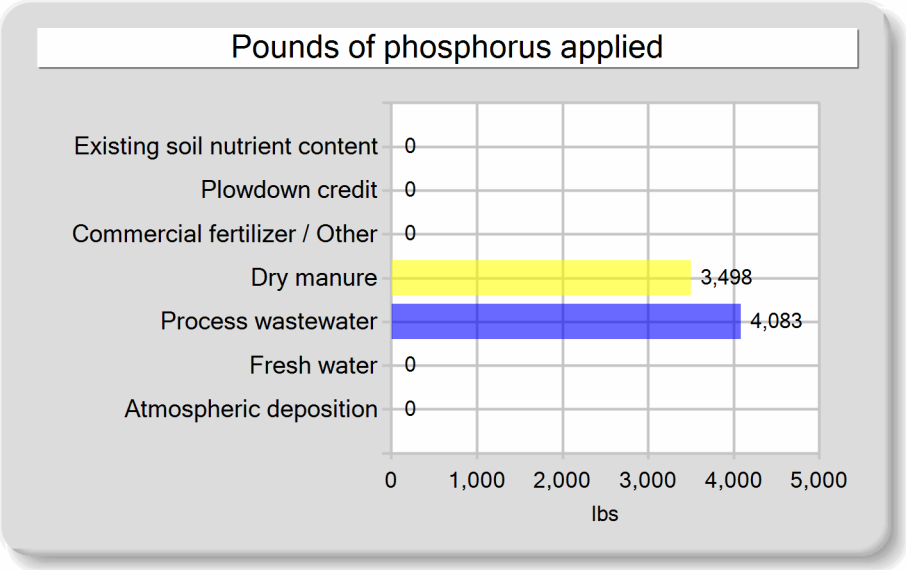
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	7,489.95	3,498.30	17,043.00	0.00
Process wastewater	16,378.18	4,082.71	55,163.12	212,006.39
Fresh water	0.00	0.00	0.00	57,721.07
Atmospheric deposition	3,948.00	0.00	0.00	0.00
Total nutrients applied	27,816.13	7,581.01	72,206.12	269,727.47
Anticipated crop nutrient removal	45,840.00	7,360.40	36,047.20	0.00
Actual crop nutrient removal	23,189.27	3,473.96	24,363.23	100,414.65
Nutrient balance	4,626.86	4,107.04	47,842.89	169,312.82
Applied to removed ratio	1.20	2.18	2.96	2.69

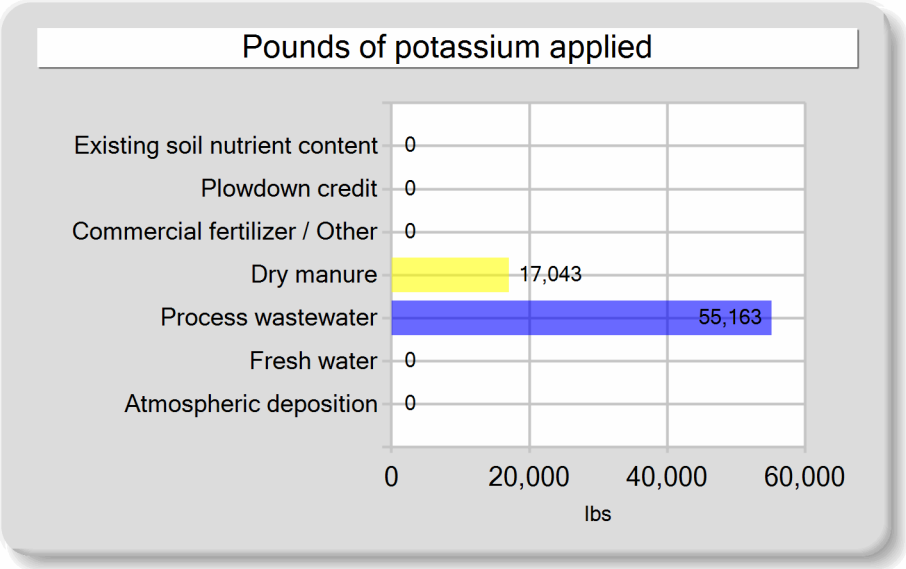
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

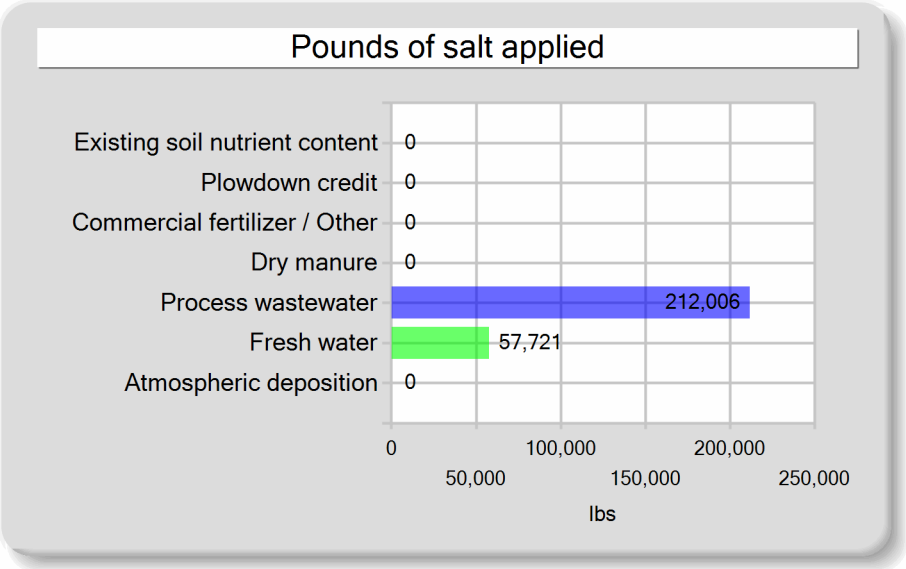


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Manuel Soares (Jun 18, 2024 16:10 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel Soares	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/18/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Manuel Soares**
Name of Dairy Facility:	**Manuel Soares & Sons Dairy**
Facility Address:	**7810 Vivian Rd, Modesto, Ca 95358**
Contact Person Name and Phone Number:	**Manuel Soares (209) 537-5692**

Manure Hauler Information:

Name of Hauling Company/Person:	**Manuel Soares & Sons Dairy**
Address of Hauling Company /Person:	**7810 Vivian Rd, Modesto, Ca 95358**
Contact Person and Phone Number:	**Manuel Soares (209) 537-5692**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Greg Nunes	**6801 Carpenter Rd**	**Modesto**	**95358**	**209-678-7933**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
3136 W. Monte Vista Ave.	**Modesto**	**95358**	**Stanislaus County**

Dates Hauled: **7/10/23 & 8/13/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____
N/A

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **1,245,000** Gallons

Method used to determine volume of process wastewater: _____
PUMP GPM x Total Run Time

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Manuel Soares (Jun 18, 2024 16:10 PDT) Date: 06/18/2024

Hauler's Signature: Manuel Soares (Jun 18, 2024 16:10 PDT) Date: 06/18/2024

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0894-01	Well #1	Ag Water	T. Madruga		11/15/2023 15:00
23K0894-02	Well #2	Ag Water	T. Madruga		11/15/2023 15:30
23K0894-03	Well #3	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: -0.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Sample Results

Sample: Well #1
23K0894-01 (Water)

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.06**	mmhos/cm	0.01	1		11/20/23 13:30	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1060**	umhos/cm	10.0	1		11/20/23 13:30	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:00	Field		BEK0553
Nitrate Nitrogen as NO3N	**24.3**	mg/L	0.1	1	10	11/16/23 22:20	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:30	SM 4500-H+	H	BEK0616
pH	**7.6**	units	1.0	1		11/20/23 13:30	SM 4500-H+	H	BEK0616

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Sample Results
(Continued)

Sample: Well #2
23K0894-02 (Water)

Sampled: 11/15/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.14**	mmhos/cm	0.01	1		11/20/23 13:31	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1140**	umhos/cm	10.0	1		11/20/23 13:31	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:30	Field		BEK0553
Nitrate Nitrogen as NO3N	**26.1**	mg/L	0.1	1	10	11/17/23 01:15	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:31	SM 4500-H+	H	BEK0616
pH	**7.8**	units	1.0	1		11/20/23 13:31	SM 4500-H+	H	BEK0616

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Sample Results
(Continued)

Sample: Well #3
23K0894-03 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.48**	mmhos/cm	0.01	1		11/20/23 13:33	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1480**	umhos/cm	10.0	1		11/20/23 13:33	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0553
Nitrate Nitrogen as NO3N	**27.5**	mg/L	0.1	1	10	11/17/23 01:37	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:33	SM 4500-H+	H	BEK0616
pH	**7.7**	units	1.0	1		11/20/23 13:33	SM 4500-H+	H	BEK0616

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0603										
Blank (BEK0603-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0603-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.2	90-110		
LCS (BEK0603-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.8	90-110		
Duplicate (BEK0603-DUP1)		**Source: 23K0892-03**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L		10.8			0.157	10
Duplicate (BEK0603-DUP2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	25.4		0.1	mg/L		24.8			2.19	10
Matrix Spike (BEK0603-MS1)		**Source: 23K0892-03**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	15.9		0.1	mg/L	5.000	10.8	101	90-110		
Matrix Spike (BEK0603-MS2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	30.4		0.1	mg/L	5.000	24.8	112	90-110		
Reference (BEK0603-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0603-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		
Reference (BEK0603-SRM3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616										
Blank (BEK0616-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0616-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.2		1.0	units						
Blank (BEK0616-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.7		1.0	units						
Duplicate (BEK0616-DUP1)		**Source: 23K0896-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.47		0.01	mmhos/cm		0.47			0.423	10
pH	7.8		1.0	units		7.8			0.255	10
Electrical Conductivity umhos	472		10.0	umhos/cm		474			0.423	10
Duplicate (BEK0616-DUP2)		**Source: 23K0899-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.65		0.01	mmhos/cm		1.64			0.298	10
Electrical Conductivity umhos	1650		10.0	umhos/cm		1640			0.298	10
pH	7.0		1.0	units		7.0			0.00	10
Reference (BEK0616-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	561			umhos/cm	538.0		104	90-110		
Reference (BEK0616-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0616-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0616-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Soares & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:02

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616 (Continued)										
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0616-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

11/16/23 15:35 23K0894

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles:

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Soares & Sons
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1	11/15/2023	1500	-.1	ND
2	Well #2	11/15/2023	1530	.2	ND
3	Well #3	11/15/2023	1600	.1	ND
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third	DLI	DLI	11.16.23 3:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0894

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice: ☑ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										